GRADE 10 MATHEMATICS - HOMEWORK WORKSHEET

NexSchool Enterprise Education Platform | Academic Session 2025-2026

---

1. Course & Assignment Metadata

Course: Mathematics (Core) | Course Code: MTH-101 | Academic Year: 2025-2026
Faculty / Instructor: Sarah Jenkins (Senior Mathematics Faculty)
Target Class: Grade 10 - Section A | Total Marks: 50 | Due Date: October 25, 2026
Topic: Unit 4 - Quadratic Equations, Polynomial Discriminants & Graph Vertex Analysis

2. General Instructions for Students

- Answer all questions in neat handwriting with complete step-by-step mathematical proofs.
- Identify the roots using factoring where possible, or use the quadratic formula x = (-b +/- sqrt(D))/(2a).
- Upload your completed homework as a PDF via the Student LMS Portal before 11:59 PM.

3. Section A: Algebraic Derivations (20 Marks)

Q1. Solve for x: 3x^2 - 14x + 8 = 0. State whether the roots are rational, irrational, or complex.
Q2. Find the discriminant D = b^2 - 4ac for the equation 2x^2 - 6x + 5 = 0 and interpret the nature of roots.
Q3. If alpha and beta are roots of x^2 - 7x + 12 = 0, evaluate alpha^2 + beta^2 and 1/alpha + 1/beta.

4. Section B: Real-World Modeling Problems (30 Marks)

Q4. A model rocket is fired vertically with initial height h(t) = -5t^2 + 40t + 10 meters.
  (a) Find the maximum height reached by the rocket using the vertex formula.
  (b) Determine the exact time in seconds when the projectile impacts the ground.
Q5. The length of a rectangular sports lawn exceeds its width by 6 meters. If its area is 216 sq meters, find the dimensions of the perimeter.

Generated electronically via NexSchool ERP Teacher Portal on Monday, October 12, 2026
Document Verification Hash: SEC-DOC-1CTBE30J